

GLM for count data

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- Gaussian: `lm`

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- Binary: `glm (family binomial / quasibinomial)`

- Gaussian: `lm`
- Binary: `glm (family binomial / quasibinomial)`
- Counts: `glm (family poisson / quasipoisson)`

- Response variable: Counts (0, 1, 2, 3...) - discrete
- Link function: $\log$

Then

$$\log(N) = a + bx$$

$$N = e^{a+bx}$$

Example dataset: Seedling counts in quadrats

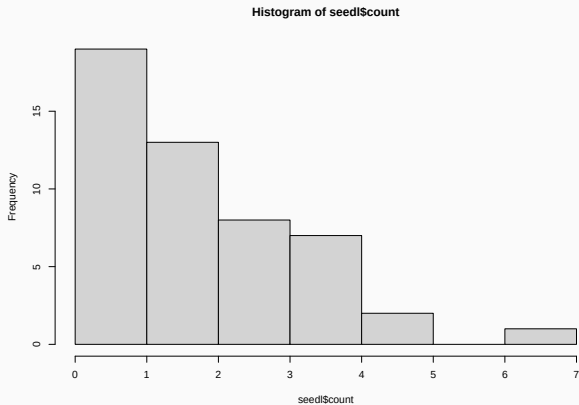
```
seedl <- read.csv('data/seedlings.csv')
```

sample	count	light	area
Min. : 1.00	Min. :0.00	Min. : 2.571	Min. :0.25
1st Qu.:13.25	1st Qu.:1.00	1st Qu.:26.879	1st Qu.:0.25
Median :25.50	Median :2.00	Median :47.493	Median :0.50
Mean :25.50	Mean :2.14	Mean :47.959	Mean :0.62
3rd Qu.:37.75	3rd Qu.:3.00	3rd Qu.:67.522	3rd Qu.:1.00
Max. :50.00	Max. :7.00	Max. :99.135	Max. :1.00

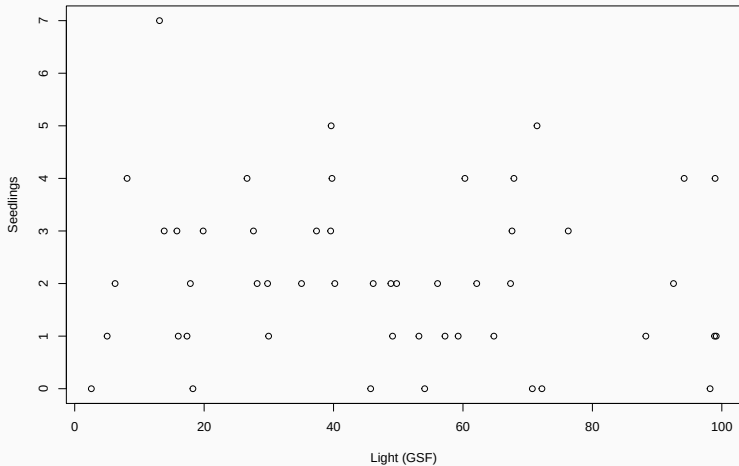
Exploring the data

```
table(seedl$count)
```

0	1	2	3	4	5	7
7	12	13	8	7	2	1



Relationship between Nseedlings and light?




```
seedl.glm <- glm(count ~ light,  
                 data = seedl,  
                 family = poisson)
```

which corresponds to

```
equatiomatic::extract_eq(seedl.glm)
```

$$\log(E(\text{count})) = \alpha + \beta_1(\text{light}) \quad (1)$$

Interpreting Poisson GLM

Call:

```
glm(formula = count ~ light, family = poisson, data = seedl)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.881805	0.188892	4.668	3.04e-06 ***
light	-0.002576	0.003528	-0.730	0.465

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

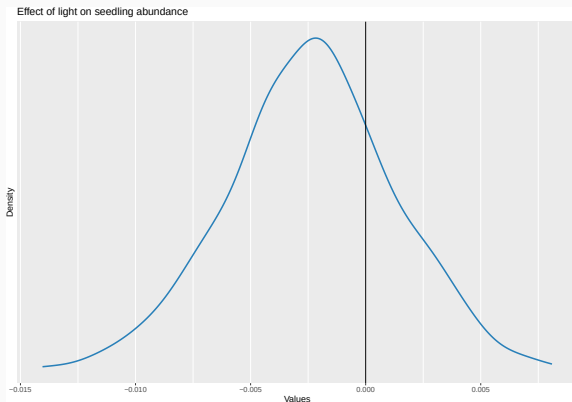
(Dispersion parameter for poisson family taken to be 1)

Null deviance: 63.029 on 49 degrees of freedom
Residual deviance: 62.492 on 48 degrees of freedom
AIC: 182.03

Number of Fisher Scoring iterations: 5

Estimated distribution of the slope parameter

```
library('parameters')  
plot(simulate_parameters(seedl.glm)) +  
  geom_vline(xintercept = 0) +  
  ggtitle('Effect of light on seedling abundance')
```



Parameter estimates are in log scale!

Parameter estimates (log scale):

```
coef(seedl.glm)[1]
```

(Intercept)

0.881805

We need to back-transform: apply the inverse of the logarithm

```
exp(coef(seedl.glm)[1])
```

(Intercept)

2.415255

Using easystats

```
library('easystats')  
parameters(seedl.glm)
```

Parameter	Log-Mean	SE	95% CI	z	p
(Intercept)	0.88	0.19	[0.50, 1.24]	4.67	< .001
light	-2.58e-03	3.53e-03	[-0.01, 0.00]	-0.73	0.465

```
parameters(seedl.glm, exponentiate = TRUE)
```

Parameter	IRR	SE	95% CI	z	p
(Intercept)	2.42	0.46	[1.65, 3.46]	4.67	< .001
light	1.00	3.52e-03	[0.99, 1.00]	-0.73	0.465

How Nseedlings decrease with light

Model-based Expectation

light	Predicted	SE	95% CI

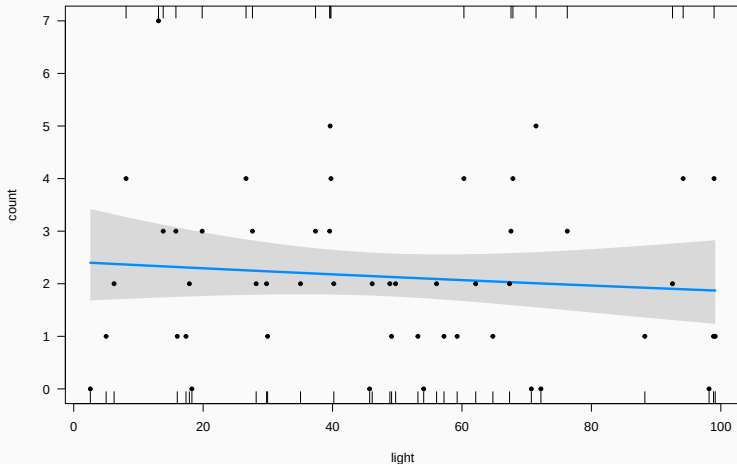
2.57	2.40	0.43	[1.68, 3.42]
13.30	2.33	0.35	[1.74, 3.13]
24.03	2.27	0.28	[1.78, 2.89]
34.76	2.21	0.23	[1.80, 2.71]
45.49	2.15	0.21	[1.78, 2.60]
56.22	2.09	0.22	[1.71, 2.56]
66.95	2.03	0.25	[1.60, 2.58]
77.68	1.98	0.29	[1.48, 2.64]
88.41	1.92	0.34	[1.36, 2.73]
99.13	1.87	0.39	[1.24, 2.83]

Variable predicted: count

Predictors modulated: light

Visualising how Nseedlings decrease with light

```
visreg(seedl.glm, scale = 'response', ylim = c(0, 7))  
points(count ~ light, data = seedl, pch = 20)
```



```
library('performance')  
r2(seedl.glm)
```

```
# R2 for Generalized Linear Regression  
Nagelkerke's R2: 0.015
```


Describing the model results

```
library('report')  
report(seedl.glm)
```

We fitted a poisson model (estimated using ML) to predict count with light (formula: count ~ light). The model's explanatory power is very weak (Nagelkerke's $R^2 = 0.01$). The model's intercept, corresponding to light = 0, is at 0.88 (95% CI [0.50, 1.24], $p < .001$). Within this model:

- The effect of light is statistically non-significant and negative (beta = $-2.58e-03$, 95% CI [$-9.57e-03$, $4.28e-03$], $p = 0.465$; Std. beta = -0.07 , 95% CI [-0.27 , 0.12])

Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald z-distribution approximation.

Model checking

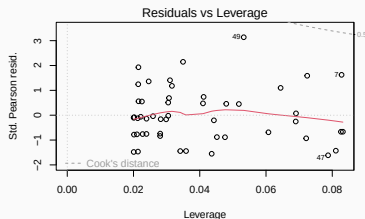
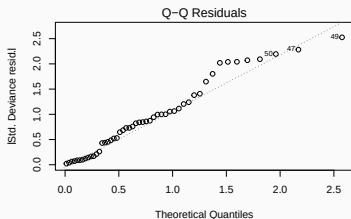
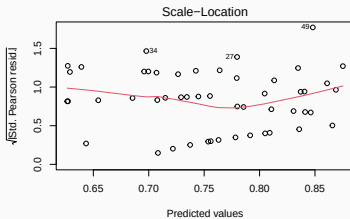
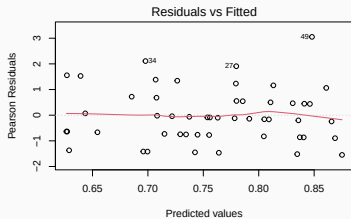
- Linearity (log response $\sim$ predictors)

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- Observations are independent

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- Mean = Variance

Checking Poisson GLM

```
plot(seedl.glm)
```



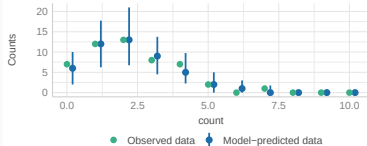
null device

Checking Poisson GLM

```
check_model(seedl.glm)
```

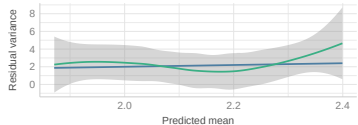
Posterior Predictive Check

Model-predicted intervals should include observed data points



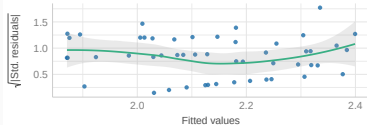
Misspecified dispersion and zero-inflation

Observed residual variance (green) should follow predicted residual variance (blue)



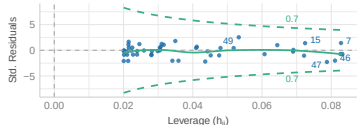
Homogeneity of Variance

Reference line should be flat and horizontal



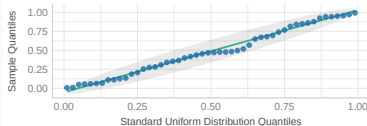
Influential Observations

Points should be inside the contour lines



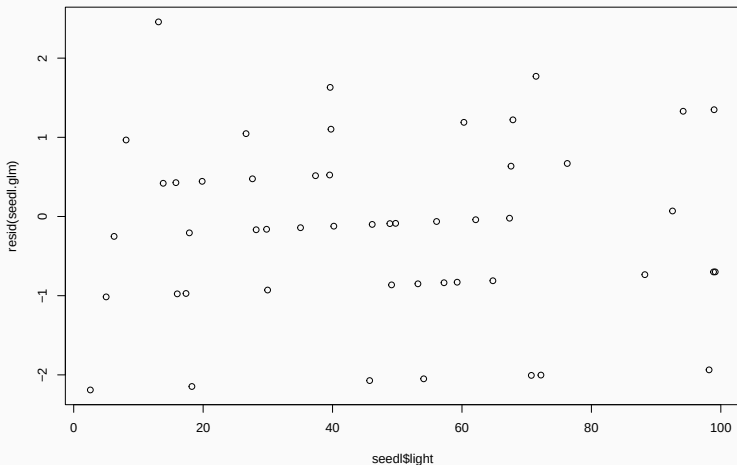
Uniformity of Residuals

Dots should fall along the line



Is there pattern of residuals along predictor?

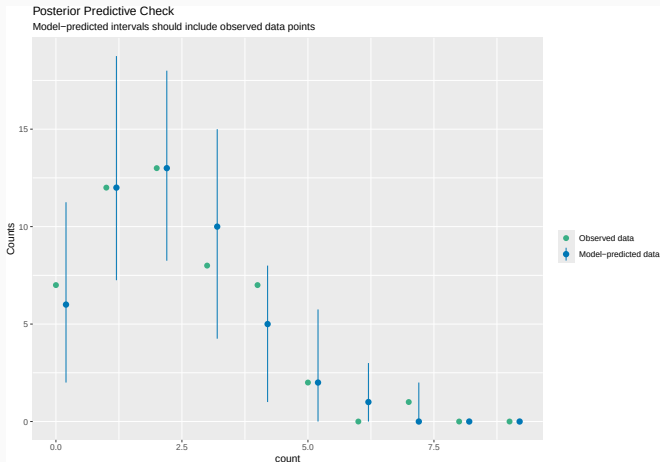
```
plot(seedl$light, resid(seedl.glm))
```



Posterior predictive checking

Simulate data from fitted model (`yrep`) and compare with observed data (`y`)

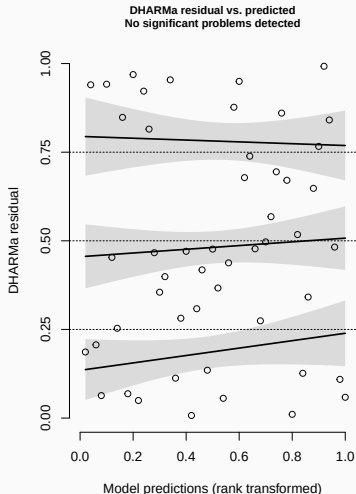
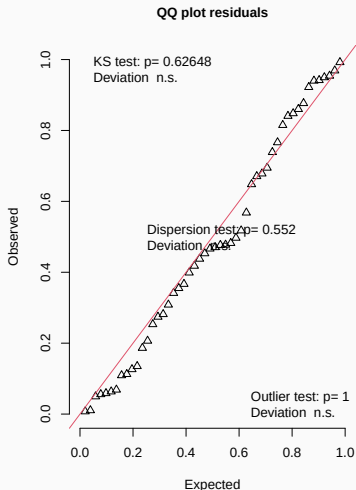
```
check_predictions(seedl.glm)
```



Residuals diagnostics with DHARMA

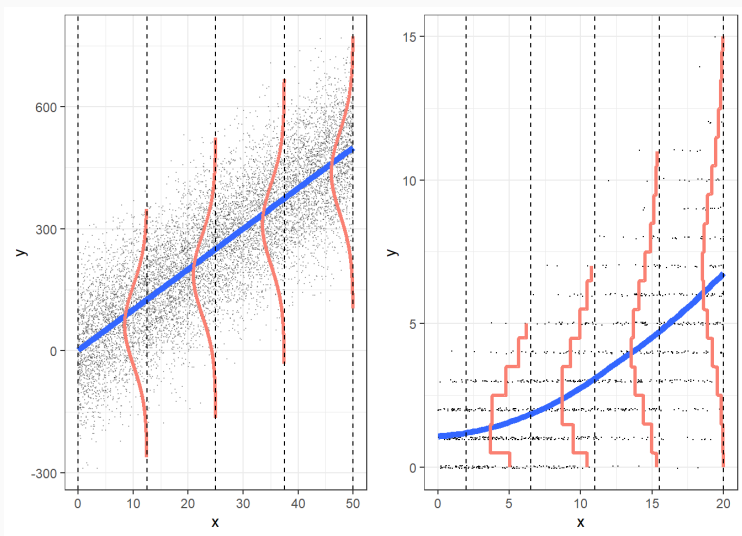
```
simulateResiduals(seedl.glm, plot = TRUE)
```

DHARMA residual



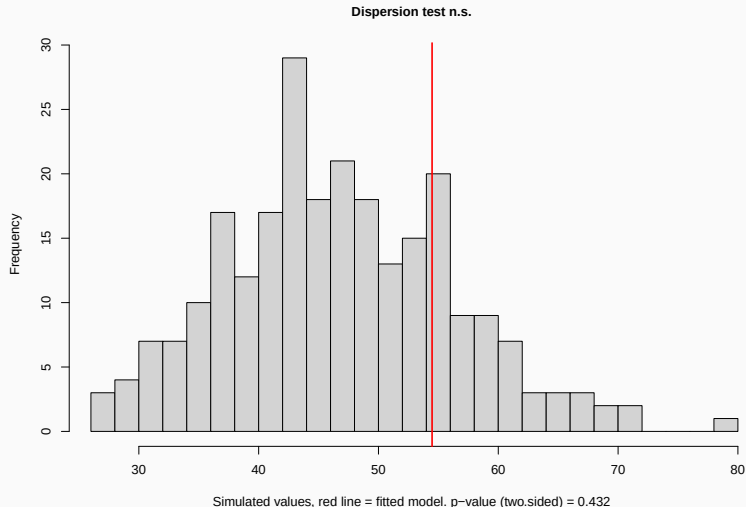
Overdispersion

Poisson GLM assumes mean = variance



Always check overdispersion with count data

```
simres <- simulateResiduals(seedl.glm, refit = TRUE)  
testDispersion(simres)
```



- Use family `quasipoisson`

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- Use negative binomial distribution (`MASS::glm.nb`)

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- Use negative binomial distribution (`MASS::glm.nb`)
- Include observation-level random effect (e.g. see [Harrison 2014](#))

Accounting for overdispersion with family quasipoisson

Call:

```
glm(formula = count ~ light, family = quasipoisson, data = seedl)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.881805	0.201230	4.382	6.37e-05 ***
light	-0.002576	0.003758	-0.685	0.496

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 1.134907)

Null deviance: 63.029 on 49 degrees of freedom
Residual deviance: 62.492 on 48 degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5

Mean estimates do not change after accounting for overdispersion

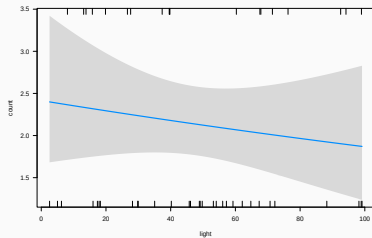
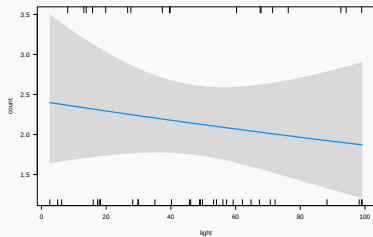
```
parameters(seedl.overdisp)
```

Parameter		Log-Mean		SE		95% CI		t(48)		p
(Intercept)		0.88		0.20		[0.47, 1.26]		4.38		< .001
light		-2.58e-03		3.76e-03		[-0.01, 0.00]		-0.69		0.493

```
parameters(seedl.glm)
```

Parameter		Log-Mean		SE		95% CI		z		p
(Intercept)		0.88		0.19		[0.50, 1.24]		4.67		< .001
light		-2.58e-03		3.53e-03		[-0.01, 0.00]		-0.73		0.465

But standard errors may change



Accounting for overdispersion using negative binomial

```
library('MASS')  
seedl.nb <- glm.nb(count ~ light, data = seedl)
```

Call:

```
glm.nb(formula = count ~ light, data = seedl, init.theta = 22.23419419,  
       link = log)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.881996	0.198213	4.450	8.6e-06 ***
light	-0.002580	0.003691	-0.699	0.485

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(22.2342) family taken to be 1)

Null deviance: 58.247 on 49 degrees of freedom
Residual deviance: 57.756 on 48 degrees of freedom
AIC: 183.83

Number of Fisher Scoring iterations: 1

Comparing Poisson and Negative Binomial

```
compare_models(seedl.glm, seedl.nb)
```

Parameter	seedl.glm	seedl.nb
(Intercept)	0.88 (0.51, 1.25)	0.88 (0.49, 1.27)
light	-2.58e-03 (-0.01, 0.00)	-2.58e-03 (-0.01, 0.00)
Observations	50	50

```
compare_performance(seedl.glm, seedl.nb)
```

Comparison of Model Performance Indices

Name	Model	AIC (weights)	AICc (weights)	BIC (weights)
seedl.glm	glm	182.0 (0.710)	182.3 (0.737)	185.9 (0.864)
seedl.nb	negbin	183.8 (0.290)	184.3 (0.263)	189.6 (0.136)

Name	Nagelkerke's R2	RMSE	Sigma	Score_log	Score_spherical
seedl.glm	0.015	1.529	1.000	-1.780	0.131
seedl.nb	0.014	1.529	1.000	-1.821	0.133

What if survey plots have
different area?

Shall we *standardise* counts dividing by sampling plot area?

Model would be: $\text{count/area} \sim \text{light}$

	sample	count	light	area
1	1	0	70.71854	0.50
2	2	1	88.26021	0.25
3	3	2	67.35133	0.50
4	4	3	67.57850	1.00
5	5	4	26.63098	0.25
6	6	3	15.79433	1.00

J. R. Statist. Soc. A (1993)
156, Part 3, pp. 379–392

Spurious Correlation and the Fallacy of the Ratio Standard Revisited

By RICHARD A. KRONMAL†

<https://doi.org/10.2307/2983064>

Use offset to account for variable sampling effort

```
seedl.offset <- glm(count ~ light,  
                    offset = log(area),  
                    data = seedl,  
                    family = poisson)
```

Note estimates now referred to area units!

Call:

```
glm(formula = count ~ light, family = poisson, data = seedl,  
     offset = log(area))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.513185	0.183245	8.258	<2e-16 ***
light	-0.005674	0.003384	-1.677	0.0936 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 95.199 on 49 degrees of freedom
Residual deviance: 92.354 on 48 degrees of freedom
AIC: 211.9

Note estimates now referred to area units!

```
exp(coef(seedl.offset)[1])
```

(Intercept)

4.541173

Prediction

Predicting number of seedlings given light

```
new.lights <- data.frame(light = c(10, 90))  
predict(seedl.glm, newdata = new.lights, type = 'response', se.fit
```

```
$fit
```

```
      1      2  
2.353841 1.915533
```

```
$se.fit
```

```
      1      2  
0.3756992 0.3502446
```

```
$residual.scale
```

```
[1] 1
```

Prediction (easystats)

```
new.lights <- data.frame(light = c(10, 90))  
estimate_expectation(seedl.glm, data = new.lights)
```

Model-based Expectation

light	Predicted	SE	95% CI
10.00	2.35	0.38	[1.72, 3.22]
90.00	1.92	0.35	[1.34, 2.74]

Variable predicted: count

```
estimate_prediction(seedl.glm, data = new.lights)
```

Model-based Prediction

light	Predicted	95% CI
10.00	2.35	[0.00, 6.00]
90.00	1.92	[0.00, 5.00]

Variable predicted: count

- Infant mortality ~ GDP

- Infant mortality ~ GDP
- Number of cones consumed by squirrels ([data](#))

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- Number of cones consumed by squirrels ([data](#))
- Elephant matings ([Poole 1989](#))